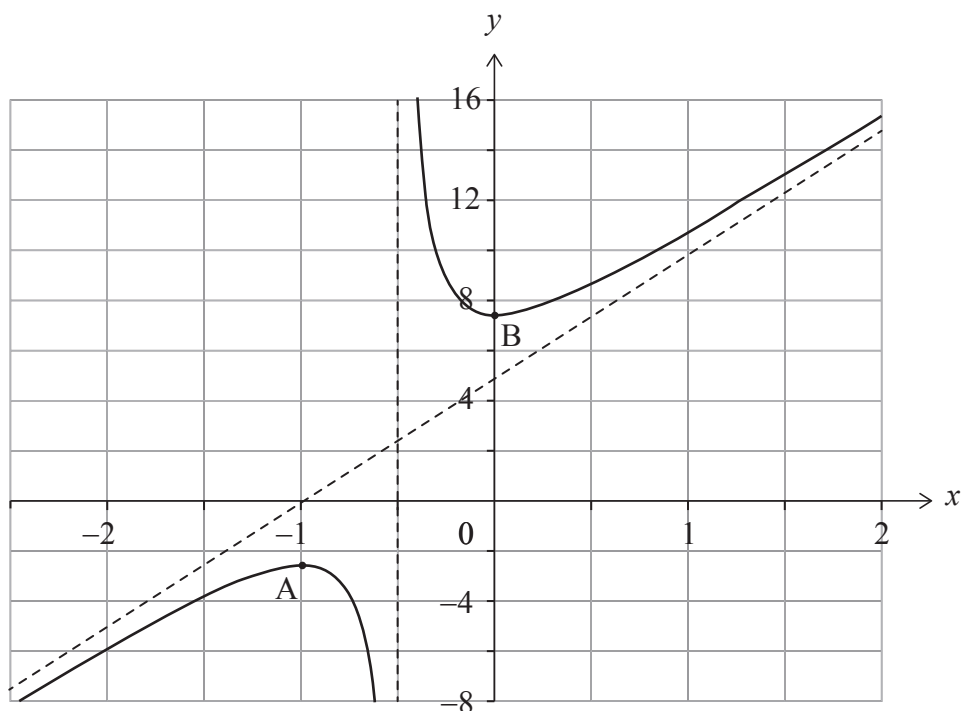


6. [Maximum mark: 7]

Consider the function f . The graph of f has a local maximum at $A\left(-1, -\frac{5}{2}\right)$, a local minimum at $B\left(0, \frac{15}{2}\right)$, a vertical asymptote at $x = -\frac{1}{2}$ and an oblique asymptote $y = 5x + 5$.

This information and part of the graph of f is shown in the following diagram.

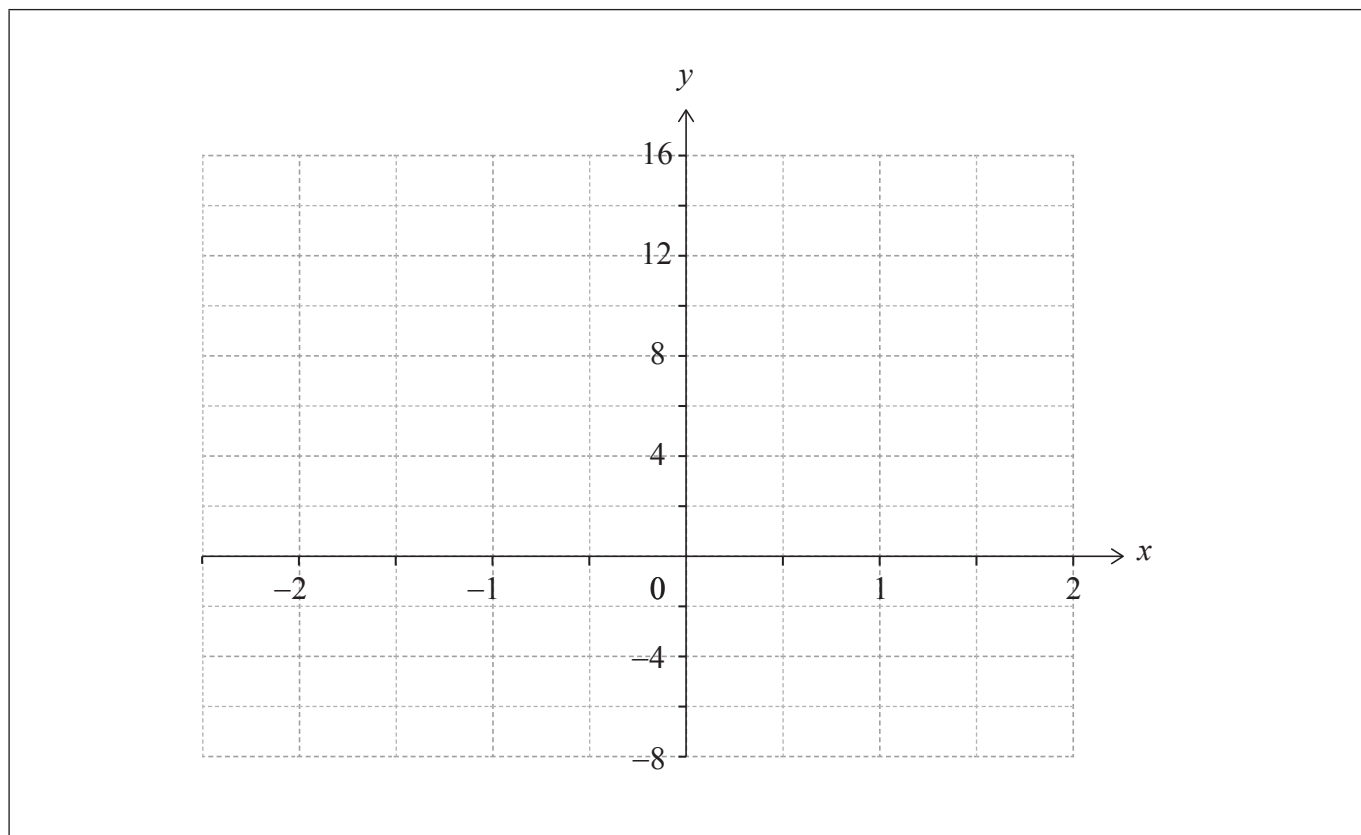


(This question continues on the following page)



(Question 6 continued)

- (a) On the following grid, sketch the graph of $y = |f(x)|$, clearly indicating any asymptotes. [4]



- (b) On the following grid, sketch the graph of $y = \frac{15}{f(x)}$, clearly indicating any asymptotes and intercepts with the axes. [3]

